Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	SWTID-2026-5878
Project Name	Automating Data Import and Relationship Mapping using Import Sets and Dot Walking
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**Employee Data Staging**	• Upload employee data (CSV/XLSX) via System Import Sets. • Automatically generate and populate the import_employee staging table.
FR-2	**Data Transformation & Mapping**	• Create a Transform Map linking import_employee to the sys_user table. • Map standard string fields (Name, Email, Phone) using Auto Map. • Configure Mapping Assist for reference fields (Department, Manager) to find matching sys_ids.
FR-3	**Transform Execution & Validation**	• Execute the Transform Map engine. • Validate successful record insertion and relationship linking in the sys_user table.
FR-4	**Incident Form Automation**	• Configure the Incident form layout. • Add dot-walked fields (Assigned to.department, Assigned to.email, Assigned to.manager) to the form view.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Incident form must auto-populate the dot-walked fields seamlessly and immediately when the "Assigned to" user is selected, without requiring a page reload.
NFR-2	**Security**	Only users with admin or designated import roles should have access to load data and configure Transform Maps. Data displayed on the Incident form must adhere to existing ServiceNow Access Control Lists (ACLs).
NFR-3	**Reliability**	The data import process should handle discrepancies gracefully. For example, if a provided manager does not exist in the system, the transform should log the error but continue processing the rest of the batch.
NFR-4	**Performance**	Processing the import sets should not cause noticeable degradation to platform performance. Dot-walked client-side lookups on the Incident form must load instantaneously.
NFR-5	**Availability**	The data import module and the modified Incident form must maintain the standard high availability of the core ServiceNow platform.
NFR-6	**Scalability**	The Transform Map logic must be capable of efficiently processing both small daily updates and large bulk imports (e.g., thousands of rows during an acquisition or major onboarding event).